

Forward and inverse kinematics of a 6-DOF robotic manipulator with a prismatic joint using MATLAB robotics toolbox

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Abstract

Robotic manipulators play a crucial role in automating industrial operations, with increasing demand in the manufacturing industry. Investigating the movement of a manipulator with a substantial number of degrees of freedom (DOF) and finding an analytical resolution to the inverse kinematics is paramount in robot modeling. This study focuses on the kinematic modeling and analysis of a 6-DOF robotic manipulator. It aims to validate the accuracy of forward and inverse kinematics calculations using the Denavit-Hartenberg (D-H) parameterization method and MATLAB GUIDE, ensuring precise motion control and path planning for high-precision applications. The 6-DOF robotic manipulator was constructed using SolidWorks, featuring five revolute joints and one prismatic joint. The D-H parameters were established for the manipulator, and kinematic equations were derived. MATLAB GUIDE was employed to perform forward and inverse kinematics calculations, and the results were validated by comparing expected and obtained values. The forward kinematics results demonstrated minimal discrepancies between expected and obtained end-effector positions, with errors ranging from 0.01 to 0.02 units. Inverse kinematics calculations also showed minor deviations in joint angles, generally within 0.01 degrees, indicating a precise match between desired and computed values. These negligible errors confirm the reliability of the D-H parameter assignment and the kinematic equations used. This study successfully simplifies the complex calculations of forward and inverse kinematics for a six-DOF robotic manipulator, providing a robust foundation for precise motion control and path planning. The findings also validate the D-H parameterization method and highlight the practical importance of accurate kinematic modeling in high-precision applications.

Keywords

Robot manipulator, Denavit-Hartenberg (D-H), Forward kinematics, Inverse kinematics, Degree of freedom.

1.Introduction

Robotic manipulators have become pivotal in automating industrial processes, significantly contributing to various sectors, including manufacturing, automotive, aerospace, and healthcare. These mechanical devices, designed to replicate the actions of a human arm, perform tasks through articulated movements with high precision, speed, and efficiency [1, 2]. The evolution from simple single-degrees of freedom (DOF) systems to sophisticated multi-DOF systems has been largely driven by the increasing demand for automation and the need for advanced manufacturing processes. As industries strive for higher productivity and quality, the role of robotic manipulators continues to expand, emphasizing the importance of their development and optimization [2, 3].

Kinematic modeling is a cornerstone of robotics, providing the mathematical framework necessary to design, control, and simulate robotic systems. It focuses on the geometric relationships and movement of robot links and joints, disregarding the forces that cause this movement. This modeling is crucial for understanding, controlling, and simulating the behavior of manipulators [4]. There are two main types of kinematic modeling: forward kinematics and inverse kinematics. Forward kinematics involves determining the position and orientation of the robot's end-effector based on given joint parameters, which is essential for defining the manipulator's reachable workspace. In contrast, inverse kinematics involves calculating the joint parameters required to achieve a specific end-effector position and orientation, presenting a more complex than the direct kinematics problem, especially as the DOF increases, and due to

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Intelligent Control of Rigid-Link Manipulators: A Systematic Review of Recent Advances and Future Trends

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ABSTRACT

As robotic manipulators increasingly operate in dynamic and safety-critical environments, the need for intelligent control strategies that ensure adaptability, robustness, and real-time performance has become critical. While earlier reviews have addressed aspects of this domain, they often lacked systematic rigor, overlooked emerging hybrid and learning-based approaches, or provided limited quantitative synthesis. The research contribution is a PRISMA-compliant systematic review of 80 peer-reviewed studies on intelligent control of rigid-link manipulators (RLMs) published between 2016 and 2024, offering both qualitative and structured comparative analysis. The methods reviewed include PID, sliding mode control (SMC), fuzzy logic, artificial neural networks (ANN), reinforcement learning (RL), genetic algorithms (GA), and hybrid combinations. Studies were assessed according to methodological clarity, experimental validation, reported performance metrics, and publication impact. A comparative summary of 25 representative studies-selected based on citation impact, methodological rigor, and diversity of control approaches-highlights performance trade-offs and strengths across techniques. The findings indicate a growing shift toward hybrid intelligent controllers, which demonstrate enhanced adaptability in addressing nonlinear dynamics and uncertainties. However, most studies remain simulation-based, with limited real-world validation and reproducibility. Major research gaps include the lack of standardized benchmarking, insufficient explainability, and limited generalizability across application domains. These insights support the development of deployable, interpretable, and reliable robotic controllers, particularly for industrial automation and medical robotics, where transparency and safety are paramount.

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1. Introduction

Rigid link manipulators (RLMs) play a critical role in industrial, medical, and service robotics due to their structural stability, mechanical simplicity, and high precision [1], [2]. However,

Comprehensive Kinematic Analysis for Optimal Performance of a 6-DOF Robotic Manipulator with Prismatic Joint (RRRRRP)

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Abstract—The research is centered on the kinematics of six-degree-of-freedom robotic manipulators, which play a crucial role in the movement of various robotic joints. The kinematics issue is a conversion between Cartesian and joint space and vice versa. The study aims to develop a mathematical model for a robotic manipulator's forward and inverse kinematics with 6-DOF, including a prismatic joint. By utilizing a Denavit–Hartenberg representation, the forward and inverse matrices were calculated. The analytical solution for the kinematics of a 6-DOF manipulator enables an exploration of the manipulator's movement between two different configurations. The robotic manipulator includes six components: five revolute joints and one prismatic joint.

Keywords: robot manipulator, direct kinematics, inverse kinematics, Denavit–Hartenberg method

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INTRODUCTION

The field of robot kinematics involves the detailed investigation of the motion of robots (kinematics). During a kinematic analysis, all links' position, velocity, and acceleration are determined without considering the forces responsible for this motion. The main point of interest in robotics research involves the development of accurate inverse kinematics models.

Two kinematic issues are examined in this paper. The forward kinematics issue involves determining the coordinates and orientation of the gripper for a specific link configuration. Conversely, the inverse kinematics problem involves determining links that align the gripper with a given position and orientation. Several methodologies have been developed to solve the inverse kinematics problem. Among them are closed-form solutions, which encompass algebraic and geometric approaches. Algebraic methods are preferred over numerical ones due to faster computational speeds and the fact that they can guarantee the existence of a solution. An algebraic solution was provided for a robotic manipulator's inverse kinematics (IK) with six degrees of freedom and a prismatic joint. There are revolute joints (represented by a “R”) and prismatic joints (represented by a “P”) in the robot's kinematic structure.

RELATED LITERATURE

The field of robot modeling and simulation has extensively investigated the subject of kinematics. Clothier et al. [1] proposed a geometric model to solve the challenge of determining the joint angles necessary for achieving autonomous positioning of a robotic system. Sahu et al. [2] derived a novel approach, known as quaternion algebra, for solving the forward kinematic problem.

Popovic et al. [3] developed a methodology for examining the arm's upper extremity motion. In another investigation, Wang et al. [4] provided a comprehensive analysis of the overall body kinematics of a radial symmetrical six-legged robot. In the paper, [5] presented a mathematical methodology for analyzing the kinematics of a humanoid robot. Cubero [6] presented an inverse kinematics (IK) model that aims to determine the values of all joint variables for a serial arm manipulator, regardless of its specific type. The foundation of this model was derived from the forward kinematic solution. Kuma [7] reported on developing and analyzing a virtual model robot, specifically a 6-DOF robotic manipulator. The study focused on establishing both forward and inverse kinematic solutions for the robot.

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СПРАВКА

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